

Multimodal Brain Tumor Segmentation Boosted by Monomodal Normal Brain Images

Huabing Liu¹, Zhengze Ni¹, Dong Nie¹, Dinggang Shen¹, *Fellow, IEEE*, Jinda Wang, and Zhenyu Tang²

Abstract—Many deep learning based methods have been proposed for brain tumor segmentation. Most studies focus on deep network internal structure to improve the segmentation accuracy, while valuable external information, such as normal brain appearance, is often ignored. Inspired by the fact that radiologists often screen lesion regions with normal appearance as reference in mind, in this paper, we propose a novel deep framework for brain tumor segmentation, where normal brain images are adopted as reference to compare with tumor brain images in a learned feature space. In this way, features at tumor regions, i.e., tumor-related features, can be highlighted and enhanced for accurate tumor segmentation. It is known that routine tumor brain images are multimodal, while normal brain images are often monomodal. This causes the feature comparison a big issue, i.e., multimodal vs. monomodal. To this end, we present a new feature alignment module (FAM) to make the feature distribution of monomodal normal brain images consistent/inconsistent with multimodal tumor brain images at normal/tumor regions, making the feature comparison effective. Both public (BraTS2022) and in-house tumor brain image datasets are used to evaluate our framework. Experimental results demonstrate that for both datasets, our framework can effectively improve the segmentation accuracy and outperforms the state-of-the-art segmentation methods. Codes are available at <https://github.com/hb-liu/Normal-Brain-Boost-Tumor-Segmentation>.

Index Terms—Brain tumor segmentation, normal brain images, normal reference, feature alignment, BraTS2022 dataset.

I. INTRODUCTION

SEGMENTATION of brain tumor from multimodal magnetic resonance (MR) images is an essential task for

subsequent diagnosis and treatment. Due to the inter-subject variance of brain tumors in location, shape, and appearance, accurate tumor segmentation is a challenging task.

With the advance of deep learning (DL), many brain tumor segmentation methods [1], [2], [3], [4], [5], [6], [7], [8] using various DL networks have been proposed, e.g., 3D convolutional neural network (CNN) [2], U-Net [9], and Transformer [10]. Although they have shown promising results, most of them try to improve the learning capacity of tumor-related features by elaborating deep network's internal structure. However, valuable external information, such as normal brain appearance, is usually ignored or underutilized. It is known that radiologists are usually first trained with normal samples before performing robust screening of lesion regions. Following this observation, anomaly detection has arisen for lesion region segmentation [11], [12], [13], [14]. The basic idea is to compare images containing lesion regions (e.g., tumor brain images) with their normal appearance reconstructed by autoencoder [15], in this way, lesion regions can be highlighted and relatively easy to be segmented. For example, Baur et al. [11] utilized a deep spatial autoencoder to reconstruct normal brain images from input tumor brain images. Then, brain tumors can be segmented using thresholding on the residual images produced by the tumor brain images and its normal appearance reconstruction. Astaraki et al. [12] proposed to use variational autoencoder (VAE) [13] to generate synthetic normal lung images. Different from [11], the residual images are concatenated with original tumor lung images and fed to the segmentation backbone. As a result, tumor regions can be highlighted in the segmentation process. Kobayashi et al. [14] proposed a discriminative network to detect brain tumor regions patch-wisely by calculating the L2 distance between tumor brain image patches and the corresponding reconstructed normal brain image patches in the learned feature space. In anomaly detection based methods, target images (with lesion regions) and reconstructed images (with normal appearance) are of the same modality and are often monomodal [16], so direct comparison (e.g., L2 distance) at image- or feature-level can be performed. However, in the context of brain tumor segmentation, routine tumor brain images are multimodal [17], e.g., T1, T1 contrast enhanced (T1c), T2, and FLAIR MR images, while normal brain images are usually monomodal, e.g., T1 MR images [18]. Therefore, existing anomaly detection based methods are difficult to be applied for multimodal brain tumor segmentation.

In this paper, we propose a novel deep learning based framework for multimodal brain tumor segmentation. In our

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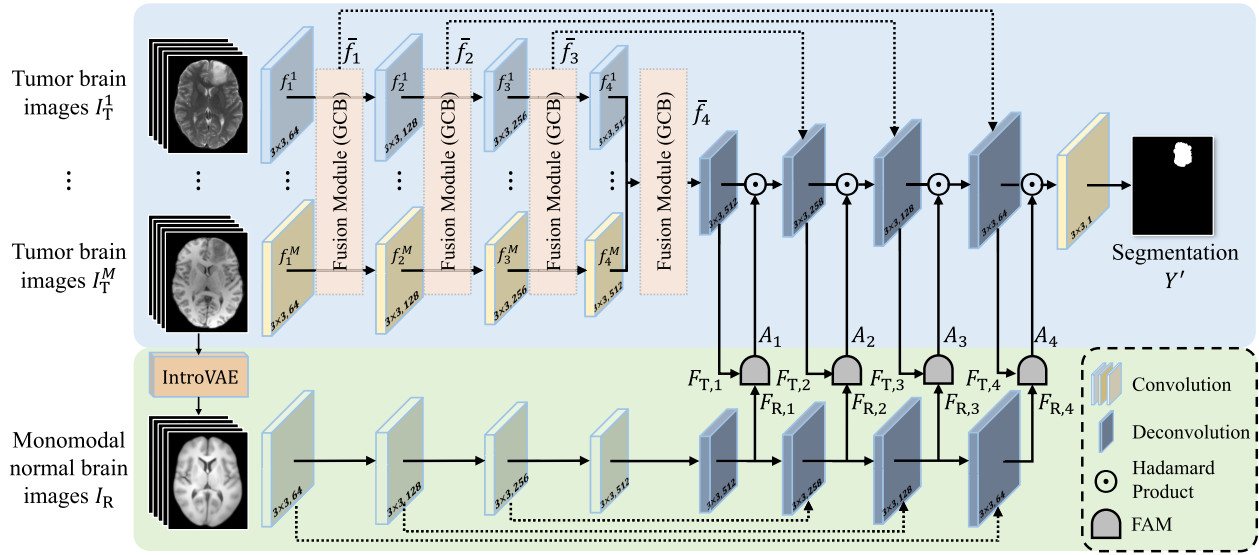


Fig. 1. Structure of the proposed framework. It is composed of two networks: the segmentation backbone (blue region), and the normal appearance network (green region). The segmentation backbone learns features from multimodal tumor brain images $\{I_T^1, \dots, I_T^M\}$ and outputs tumor segmentation results Y' . A global correlation block (GCB) is used in the segmentation backbone to fuse features learned from different modalities. The normal appearance network learns features from monomodal normal brain images I_R reconstructed by introspective variational autoencoder (IntroVAE). At each decoding level, features from both networks are sent to the feature alignment module (FAM) to perform feature alignment and comparison to generate the attention map, by which tumor-related features in the segmentation backbone can be highlighted and enhanced.

framework, a new self-attention based global correlation block (GCB) is proposed to achieve effective fusion of tumor brain features learned from multimodal tumor brain images. More importantly, monomodal normal brain images are adopted in our framework as reference, from which normal brain features are learned and compared with the tumor brain features. In this way, tumor-related features can be highlighted and enhanced, and the segmentation performance can be improved. To solve the incomparable issue between features learned from multimodal and monomodal images, a new feature alignment module (FAM) is presented. Based on FAM, the distribution of normal brain features is consistent/inconsistent with the tumor brain features at normal/tumor regions, facilitating the feature comparison as well as the final segmentation. The proposed framework is evaluated using public (BraTS2022) and in-house tumor brain image datasets. The experimental results demonstrate that our framework outperforms the state-of-the-art segmentation methods in both datasets. Our contributions are summarized as follows:

- A novel multimodal deep learning framework for brain tumor segmentation is proposed, where monomodal normal brain images are adopted as reference and compared with multimodal tumor brain images in a learned feature space, by which tumor-related features can be highlighted and enhanced.
- A new feature alignment module (FAM) is presented to solve the incomparable issue between features learned from multimodal tumor brain images and monomodal normal brain images, making the feature comparison more effective.
- Evaluation results of public (BraTS2022) and in-house datasets show superior performance of our framework over the state-of-the-art segmentation methods. Moreover, further ablation studies confirm the effectiveness of the proposed components.

- To the best of our knowledge, our framework is the first work, which presents a possible way of using normal brain images to boost tumor brain image segmentation in the context of clinical situation, i.e., tumor brain images and normal brain images are of different number of modalities.

II. METHOD

The structure of our framework is shown in Fig. 1. The framework is composed of two networks: the segmentation backbone and the normal appearance network. The input of the segmentation backbone is multimodal tumor brain images denoted as $I_T^1, \dots, I_T^M$, where M indicates the number of different modalities. The input of the normal appearance network is monomodal normal brain images I_R generated by introspective variational autoencoder (IntroVAE) [19] from the T1 sequence of the multimodal tumor brain images, e.g., I_T^M . Considering high computational efficiency of 2D convolution and the advantage of using 3D spatial information, all inputs of our framework are 2.5D slices, which contain one 2D slice to be segmented and its K neighboring slices, i.e., $2K+1$ number of slices.

In our framework, the segmentation backbone learns features from the input multimodal tumor brain images and produces the final tumor segmentation results. The normal appearance network learns features from the input monomodal normal brain images, and these features are sent to the segmentation backbone to highlight and enhance the tumor-related features via feature comparison. In this way, the final tumor segmentation results can be improved. Details of each network are presented in the following sections.

A. The Segmentation Backbone

The segmentation backbone is of encoder-decoder structure, and skip connections are adopted to combine low-level

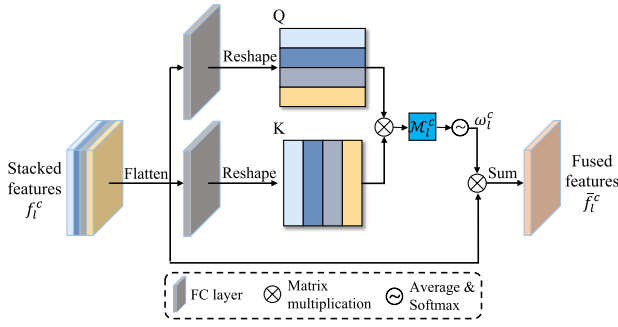


Fig. 2. Global correlation block (GCB) at the l -th encoding level. The input f_l^c is formed by features at the c -th channel in f_l^m , $m = 1, \dots, M$. Two parallel fully connected layers are used to get Q and K , based on which the correlation matrix $\mathcal{M}_l^c$ is obtained and used to get the fused feature $\bar{f}_l^c$.

and high-level features. Considering that early fusion of features learned from different modalities could lose modality-specific information [20], [21], [22], multi-encoder structure is adopted, where each encoding pathway is responsible for learning features from one image modality of the tumor brain. There are four convolution layers in each encoding pathway, and each convolution layer uses 3×3 kernel with stride of two followed by batch normalization (BN) and rectified linear unit (ReLU) activation. The output features of the first convolution layer have 64 channels, and the channel number is doubled after each subsequent convolution layers.

To fully exploit correlations among features from different modalities and perform effective multimodal image feature fusion before sending to the decoder via the skip connections, inspired by cross modality convolution (CMC) [23] and self-attention [10], [24], we propose a global correlation block (GCB) for the fusion module (see Fig. 2).

1) *The Fusion Module With Global Correlation Block (GCB)*: At each encoding level l of the backbone, the inputs of the fusion module are features $f_l^m \in \mathbb{R}^{H \times W \times C}$, learned from tumor brain images of different modalities I_T^m , $m = 1, \dots, M$, respectively. Then features at the same channel in f_l^m are concatenated, getting C number of features $f_l^c \in \mathbb{R}^{H \times W \times M}$, $c = 1, \dots, C$. In CMC, each f_l^c is proceeded by 3D convolution with kernel size of $M \times 1 \times 1$ and concatenated together to get the final fused features. We notice that such convolution has limited receptive fields and lacks global correlations. To this end, in GCB, a self-attention mechanism is adopted to explore the global correlations among all modalities. Specifically, f_l^c is flattened to $M \times (H \times W)$ and sent to two parallel fully connected (FC) layers. The resulting two features are reshaped as $Q \in \mathbb{R}^{M \times d}$ and $K \in \mathbb{R}^{d \times M}$ respectively to get the correlation matrix $\mathcal{M}_l^c$ defined as:

$$\mathcal{M}_l^c = \beta QK, \quad (1)$$

where β is used to counteract the numerical explosion. Similar to [10], β is set to $\frac{1}{\sqrt{d}}$, and d is set to 768 as adopted in [25]. In $\mathcal{M}_l^c$, each element $\mathcal{M}_l^c(i, j)$ indicates the correlation between the i -th and j -th modalities. Based on $\mathcal{M}_l^c$, the weighting vector $\omega_l^c = \{\omega_l^c(m) | m = 1, \dots, M\}$ is calculated by:

$$\omega_l^c(m) = \frac{e^{\sum_{j=1}^M \mathcal{M}_l^c(m, j)/M}}{\sum_{i=1}^M e^{\sum_{j=1}^M \mathcal{M}_l^c(i, j)/M}}, \quad (2)$$

where $\omega_l^c(m)$ is the weight for the m -th channel (i.e., modality) in f_l^c . In this way, the fused feature of channel c , i.e., $\bar{f}_l^c$, can be obtained by:

$$\bar{f}_l^c = \sum_{m=1}^M \omega_l^c(m) \cdot f_l^c(m), \quad (3)$$

where $f_l^c(m)$ is the feature at m -th channel in f_l^c . At last, the fused features of all channels $\bar{f}_l^c$, $c = 1, \dots, C$ are concatenated as the final fused feature $\bar{f}_l$ at the l -th encoding level of the segmentation backbone. Since the global correlations are considered in GCB, the feature fusion is more effective and comprehensive as compared with CMC.

2) *The Decoding Stage of Segmentation Backbone*: The decoding stage contains four deconvolution layers and one convolution layer. The input of the first deconvolution layer, i.e., decoding level $l = 1$, is the fused feature produced by the fusion module at the bottleneck of the encoder (i.e., $\bar{f}_4$), while for the rest deconvolution layers ($l = 2, 3, 4$), the input includes feature from the previous deconvolution layer enhanced by the attention map A_l (i.e., $F_{T, l-1} \odot A_{l-1}$) and the fused feature (i.e., $\bar{f}_{5-l}$) from the corresponding fusion module at the encoding stage. The attention map A_l is produced by the feature alignment module (FAM), which is discussed later. A softmax nonlinear unit is applied after the last convolution layer to generate the final segmentation result Y' , which is in the form of a probability map, and the value at each channel in Y' indicates the probability of each tumor region of interest (ROI).

The Dice index loss [26] is adopted as the loss function of the segmentation backbone, which is defined as:

$$\mathcal{L}_{\text{Dice}}(Y, Y') = -\frac{2 \sum_{x \in \Omega} Y(x) \cdot Y'(x)}{\sum_{x \in \Omega} Y(x) + \sum_{x \in \Omega} Y'(x)}, \quad (4)$$

where x is the position at image region Ω , Y' and Y are the segmentation result and the ground truth, respectively.

B. The Normal Appearance Network

The normal appearance network is a simple U-Net. It is used to learn features from monomodal normal brain images. Considering commonly used image modality for normal brain, the modality of input monomodal normal brain images I_R is set to T1. Inspired by anomaly detection methods, in our framework, I_R is reconstructed from the T1 MR tumor brain image (e.g., I_T^M) contained in the input multimodal tumor brain images using IntroVAE (see Fig. 1).

1) *The IntroVAE*: As shown in Fig. 3a, IntroVAE is composed of an encoder for projecting input brain images to a latent distribution and a decoder for producing reconstructed brain images based on the latent distribution. IntroVAE is trained in an adversarial manner by self-estimating the differences between input and reconstructed normal brain images. Given N T1 MR normal brain images from healthy subjects (denoted as I_H^i , $i = 1, \dots, N$) as training images, the encoder projects each normal brain image I_H^i to a feature embedding $z_H = \mu + \epsilon \cdot \sigma$, where μ and σ are mean and deviation of z estimated by the encoder, respectively, and ϵ is a randomly generated vector under the standard normal distribution. Based

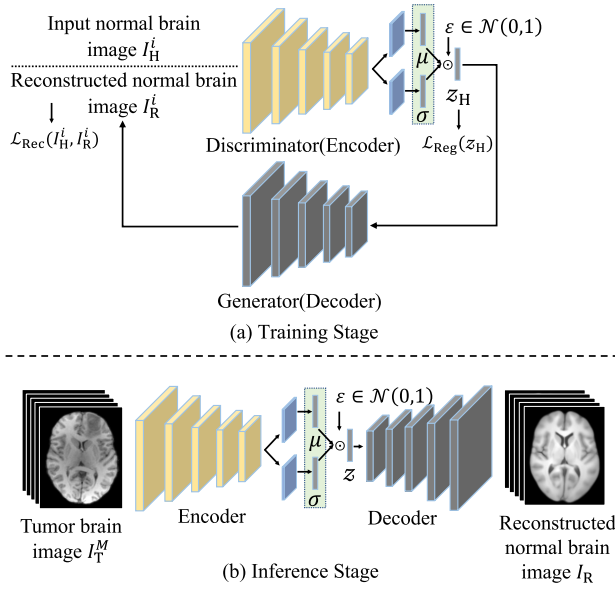


Fig. 3. Structure of IntroVAE. The encoder is trained to maximize $\mathcal{L}_{Reg}$ of the reconstructed normal brain image to discriminate it from the real one, while the decoder is responsible for reconstructing the normal brain image with small $\mathcal{L}_{Rec}$, by which $\mathcal{L}_{Reg}$ can be minimized to confuse the encoder. In the inference stage, T1 MR tumor brain image I_T^M is used as input of IntroVAE to reconstruct the normal brain image I_R .

on z_H , the decoder can produce the corresponding reconstructed normal brain image I_R^i . A reconstruction loss $\mathcal{L}_{Rec}$ is applied on I_H^i and I_R^i to minimize the reconstruction error, and a regularization loss $\mathcal{L}_{Reg}$ is applied on z_H to enforce the posterior distribution to match the prior distribution. In this way, effective representation of normal brain appearance can be achieved. Details of IntroVAE are referred to [19].

After the training of IntroVAE, T1 sequence, e.g., I_T^M , selected from the multimodal tumor brain images $I_T^1, \dots, I_T^M$ is used as the input of IntroVAE, where I_T^M is projected to a certain feature embedding z representing the closest normal brain appearance. Based on the feature embedding, the corresponding T1 MR normal brain image I_R can be reconstructed (see Fig. 3b) and used in the normal appearance network to learn normal brain features. The normal brain features are sent to the segmentation backbone as reference, and by comparing them with the tumor brain features, tumor-related features can be highlighted and enhanced for accurate tumor segmentation. However, in the context of brain tumor segmentation, the normal brain features learned from monomodality and the tumor brain features from multimodality are incomparable. To solve this issue, a new feature alignment module is proposed.

2) *The Feature Alignment Module*: We denote the tumor and normal brain features at each decoding level l of the segmentation backbone and the normal appearance network as $F_{T,l}$ and $F_{R,l}$, $l = 1, \dots, 4$, respectively (see Fig. 1). $F_{T,l}(x)$ and $F_{R,l}(x)$ are feature at position x . The main idea of FAM is to make the distribution of $F_{R,l}(x)$ consistent/inconsistent with $F_{T,l}(x)$ at normal/tumor regions. In this way, tumor regions in $F_{T,l}$ can be effectively highlighted according to feature consistency compared with the reference $F_{R,l}$. Structure of FAM is shown in Fig. 4. It is composed of two parts: the first part is feature alignment, where two 1×1 convolution layers ζ and g are adopted to align $F_{T,l}$ and $F_{R,l}$ at normal regions;

and the second part is feature comparison, where regions with low feature consistency between $\zeta(F_{T,l})$ and $g(F_{R,l})$, e.g., $\zeta(F_{T,l})(x)$ at tumor regions vs. $g(F_{R,l})(x)$ at normal regions, are identified to produce the attention map A_l for highlighting and enhancing the tumor regions in $F_{T,l}$. Details of each part are discussed as follows.

In the feature alignment part as shown in Fig. 4a, $\zeta(F_{T,l})$ and $g(F_{R,l})$ should have similar/different feature distribution at normal/tumor regions. To make the training of ζ and g more effective, a representation learning method called Simple Siamese network (SimSiam) [27] is adopted (see Fig. 4c). The SimSiam network is composed of an encoder $\mathcal{F}$, a projection MLP $\mathcal{K}$, and a prediction MLP $\mathcal{H}$. It takes each positive sample pair $\{I_1, I_2\}$, i.e., samples of the same class, as input and maximizes the cosine similarity between two output vectors p_1 and v_2 :

$$D(I_1, I_2) = -\frac{p_1}{\|p_1\|_2} \cdot \frac{v_2}{\|v_2\|_2} \quad (5)$$

where $p_1 \triangleq \mathcal{H}(\mathcal{K}(\mathcal{F}(I_1)))$, $v_2 \triangleq \mathcal{K}(\mathcal{F}(I_2))$, and $\|\cdot\|_2$ is L2-norm. In our framework, the feature pairs $\{\zeta(F_{T,l})(x), g(F_{R,l})(x)\}$, $x \in \Omega_{NR}$ (x at normal region) are defined as positive samples, and the normal region Ω_{NR} is determined according to the label images in the training dataset. In FAM, ζ and g are regarded as the encoder $\mathcal{F}$ in the SimSiam network, and the loss function of the SimSiam network is defined as:

$$\mathcal{L}_{Sim}(F_{T,l}, F_{R,l}) = \frac{1}{2} \sum_{x \in \Omega_{NR}} D(F_{T,l}(x), F_{R,l}(x)) + D(F_{R,l}(x), F_{T,l}(x)). \quad (6)$$

It is worth noting that the SimSiam network is used in the training stage and is removed at the inference stage. Moreover, to make the segmentation backbone focus on tumor segmentation rather than feature alignment, a stop gradient operation is applied before ζ to stop back-propagation to the segmentation backbone (see Fig. 4a).

In the feature comparison part as shown in Fig. 4b, feature consistency between $\zeta(F_{T,l})$ and $g(F_{R,l})$ is measured to get the attention map A_l using soft-attention [28], [29], which is defined as:

$$A_l = \sigma_2(\psi(\sigma_1(-\zeta(F_{T,l}) \cdot g(F_{R,l})))) \quad (7)$$

where σ_1 is ReLU, σ_2 is sigmoid activation function. The attention map A_l is used to highlight and enhance tumor-related features in $F_{T,l}$, i.e., $F_{T,l} \odot A_l$.

FAM is applied at each level of the decoding stage in the segmentation backbone. In this way, the tumor-related features could be strengthened in a coarse-to-fine manner, and the final tumor segmentation results can be improved.

III. RESULTS

The proposed framework is evaluated using a public dataset BraTS2022 [30], which contains multimodal MR tumor brain images (T1, T1c, T2 and FLAIR) of 1251 glioma patients. Corresponding manually labeled tumor masks of three tumor ROIs, i.e., edema, enhancing core, and necrosis, are available.

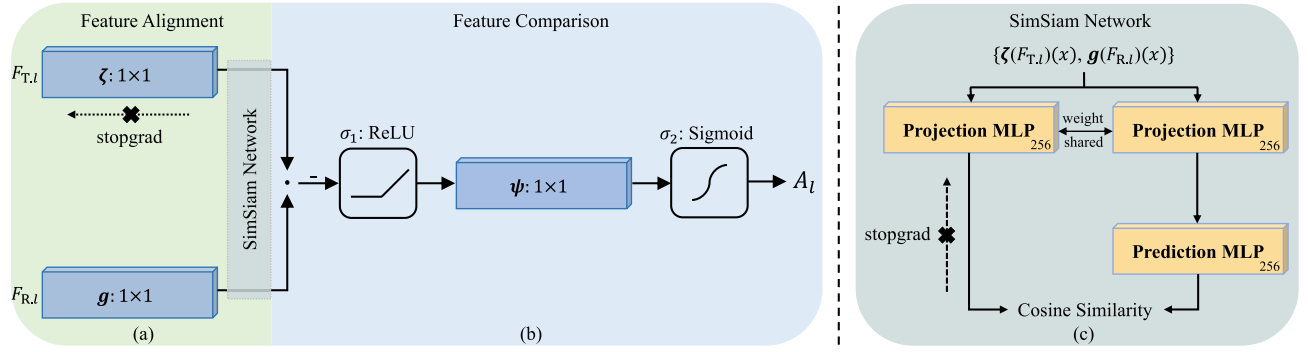


Fig. 4. Structure of FAM at decoding level l . In the feature alignment part, two 1×1 convolutional layers (ζ and g) are adopted to align tumor brain features $F_{T,l}$ and normal brain features $F_{R,l}$, and the feature consistency at normal regions is improved by the SimSiam network during training. In the feature comparison part, the attention map A_l is generated in a dot-product soft-attention manner.

TABLE I
SUMMARY OF THE GBM PATIENTS IN THE IN-HOUSE DATASET

Age	Gender	Location	Hemisphere
51.6 ± 14.6	36 Female 68 Male	23 Parietal lobe	48 Left 56 Right
		55 Frontal lobe	
		41 Temporal lobe	
		13 Occipital lobe	
		10 Basal ganglia	
		10 Corpus callosum	

Besides the BraTS2022 dataset, an in-house dataset captured by our cooperation hospital is also adopted in the experiment. The in-house dataset contains multimodal MR tumor brain images, including T1c, B0, mean diffusivity (MD), and fractional anisotropy (FA) of 104 glioblastoma (GBM) patients. The latter three modalities are derived from diffusion-weighted imaging (DWI). The key statistics of the GBM patients in the in-house dataset are summarized in Table I, including age, sex, tumor location, and the located brain hemisphere. Manually labeled whole tumor masks are available in the in-house dataset. Although the in-house dataset has no T1 modality, considering that T1c is T1 with contrast agents in the vessel and both modalities exhibit similar appearance of gray matter and white matter at normal regions [31], for the in-house dataset, we adopt T1c MR tumor brain images as the input of IntroVAE to reconstruct T1 MR normal brain images.

It is worth noting that IntroVAE is trained independently before integrated into our framework, i.e., the parameters of IntroVAE are fixed in the training stage of our framework. Specifically, the IXI dataset [32], which contains 581 T1 MR normal brain images, are used to train IntroVAE. All images of the IXI dataset go through the same pre-processing stage used in the BraTS2022 and in-house datasets to reduce possible distribution shift. The training strategy of IntroVAE follows the paper [14], where the hyperparameters, such as the batch size and maximal number of training epochs, are set to 120 and 200, respectively.

Besides our framework, existing state-of-the-art methods, including V-Net [33], Attention U-Net [34], nnU-Net [35], UNETR [36], nnFormer [37], TransBTS [5], MultiCNN [38], MultiFormer [39], and TuningUNet [40] are also evaluated. Moreover, ablation experiments of our framework are also conducted. Specifically, Baseline-1 and Baseline-2 use only

the segmentation backbone of our framework (no normal appearance network), and the only difference between them is that Baseline-1 uses cross modality convolution (CMC) in the fusion module, while global correlation block (GCB) is adopted in Baseline-2. Baseline-3 is similar to our framework, but it does not use SimSiam network in the feature alignment module (FAM).

For both public and in-house datasets, we randomly split all samples into 70%, 10% and 20% for training, validation and testing, respectively. All methods under evaluation are implemented using PyTorch and trained with RTX 3090 GPU. The input of all methods under evaluation is 2.5D slices with $K = 2$, i.e., five successive slices. The batch size is set to four, and the maximal number of training epochs is 300.

A. Evaluation of Segmentation Results

Accuracy of tumor segmentation results is quantified using Dice score, Sensitivity, Hausdorff distance, Precision, Specificity, and Jaccard index. Specifically, the Dice score and Jaccard index are used to evaluate the overlap ratio between the segmentation results and the ground truth, the Hausdorff distance measures the consistency between the boundaries of the segmented tumor and the ground truth, the Sensitivity is an indicator of the true positive rate, while the Precision and Specificity can indicate the segmentation ability to correctly identify the tumor and non-tumor regions.

For the BraTS2022 dataset, the average and standard deviation of the six metrics on enhancing tumor (ET, i.e., enhancing core), tumor core (TC, i.e., ET+necrosis), and whole tumor (WT, i.e., ET+TC+edema) are calculated, while for the in-house dataset, the corresponding metrics on whole tumor are calculated. Details of the evaluation results comparing with the nine state-of-the-art methods and the ablation study are shown in Table II. It is clear that our framework outperforms the other methods in terms of all metrics on both datasets. Moreover, the evaluation results of the ablation study demonstrate that each proposed component in our framework, including GCB, normal appearance network, and SimSiam based FAM, plays an effective and positive role in the segmentation. The patient-wise Wilcoxon signed rank test [41] is adopted to the evaluation results between our framework and the other methods, and all resulting p values are < 0.05 , i.e., statistical significance. It is worth noting that although Baseline-3 adopts

TABLE II
EVALUATION RESULTS OF ALL METHODS UNDER COMPARISON

Dataset	Method		Dice score	Jaccard index	HD	Sensitivity	Precision	Specificity
BraTS 2022	V-Net [33]	ET	0.797±0.194	0.696±0.212	10.694±19.135	0.849±0.195	0.792±0.196	0.992±0.002
		TC	0.814±0.221	0.731±0.242	13.674±20.565	0.839±0.239	0.841±0.187	0.994±0.003
		WT	0.877±0.108	0.794±0.138	13.463±19.383	0.881±0.137	0.888±0.104	0.997±0.005
	Atten. U-Net [34]	ET	0.814±0.192	0.720±0.207	14.498±10.252	0.845±0.198	0.826±0.178	0.992±0.003
		TC	0.836±0.192	0.758±0.226	10.252±16.500	0.853±0.216	0.859±0.175	0.994±0.004
		WT	0.889±0.103	0.813±0.133	12.511±19.155	0.903±0.125	0.890±0.105	0.997±0.003
	nnU-Net [35]	ET	0.829±0.167	0.751±0.254	3.582±6.597	0.819±0.255	0.875±0.140	0.994±0.005
		TC	0.875±0.136	0.813±0.247	5.219±10.210	0.856±0.243	0.917±0.122	0.996±0.006
		WT	0.919±0.060	0.858±0.113	5.974±13.016	0.899±0.113	0.939±0.062	0.996±0.003
	UNETR [36]	ET	0.834±0.185	0.748±0.206	6.915±13.715	0.842±0.199	0.862±0.168	0.992±0.004
		TC	0.847±0.206	0.776±0.228	9.609±15.458	0.846±0.214	0.890±0.166	0.995±0.004
		WT	0.902±0.099	0.834±0.132	10.454±18.364	0.896±0.121	0.918±0.097	0.997±0.005
	nnFormer [37]	ET	0.842±0.169	0.756±0.195	5.429±10.275	0.848±0.185	0.867±0.157	0.992±0.002
		TC	0.859±0.186	0.789±0.215	10.089±17.450	0.870±0.188	0.883±0.167	0.995±0.002
		WT	0.909±0.828	0.843±0.118	9.427±16.778	0.906±0.110	0.922±0.795	0.996±0.003
	TransBTS [5]	ET	0.850±0.177	0.770±0.200	4.330±9.708	0.844±0.201	0.892±0.139	0.994±0.003
		TC	0.873±0.189	0.810±0.214	6.043±11.971	0.859±0.205	0.921±0.146	0.996±0.005
		WT	0.913±0.081	0.849±0.115	9.096±16.448	0.913±0.104	0.921±0.079	0.996±0.004
	MultiCNN [38]	ET	0.852±0.176	0.772±0.199	3.789±9.510	0.847±0.197	0.882±0.140	0.994±0.003
		TC	0.873±0.191	0.811±0.216	5.122±10.265	0.861±0.202	0.916±0.124	0.994±0.005
		WT	0.915±0.082	0.852±0.115	7.420±13.502	0.902±0.110	0.937±0.066	0.997±0.007
	MultiFormer [39]	ET	0.843±0.181	0.760±0.204	5.060±11.903	0.839±0.202	0.882±0.153	0.994±0.002
		TC	0.856±0.196	0.786±0.223	7.383±13.025	0.848±0.210	0.908±0.149	0.995±0.003
		WT	0.911±0.087	0.846±0.120	8.820±16.547	0.904±0.115	0.927±0.078	0.997±0.005
	TuningUNet [40]	ET	0.857±0.167	0.777±0.192	3.519±6.246	0.860±0.181	0.883±0.154	0.994±0.002
		TC	0.875±0.179	0.811±0.208	5.831±9.994	0.869±0.191	0.912±0.143	0.996±0.003
		WT	0.919±0.073	0.859±0.110	5.920±11.229	0.907±0.107	0.942±0.057	0.997±0.003
	Baseline-1	ET	0.838±0.172	0.753±0.198	5.147±12.397	0.837±0.188	0.866±0.164	0.997±0.004
		TC	0.870±0.167	0.802±0.201	7.833±15.095	0.864±0.180	0.900±0.154	0.996±0.006
		WT	0.892±0.105	0.823±0.136	8.135±14.198	0.879±0.141	0.919±0.089	0.995±0.007
	Baseline-2	ET	0.848±0.172	0.764±0.195	4.540±10.720	0.846±0.189	0.880±0.151	0.997±0.003
		TC	0.879±0.160	0.811±0.196	7.007±13.272	0.867±0.175	0.917±0.136	0.997±0.005
		WT	0.908±0.091	0.841±0.121	8.027±13.155	0.894±0.122	0.932±0.083	0.995±0.006
	Baseline-3	ET	0.850±0.160	0.765±0.188	4.445±9.487	0.858±0.180	0.871±0.143	0.997±0.003
		TC	0.880±0.156	0.813±0.192	6.696±11.982	0.880±0.169	0.907±0.130	0.997±0.004
		WT	0.908±0.090	0.841±0.122	7.208±11.894	0.898±0.122	0.928±0.077	0.995±0.006
	Ours	ET	0.869±0.166	0.780±0.191	3.323±5.785	0.865±0.186	0.893±0.139	0.998±0.004
		TC	0.897±0.134	0.834±0.175	5.118±10.104	0.898±0.147	0.921±0.139	0.998±0.004
		WT	0.929±0.066	0.868±0.099	5.844±10.469	0.914±0.100	0.945±0.052	0.997±0.003
In-house	V-Net [33]		0.853±0.050	0.696±0.148	5.850±1.768	0.832±0.084	0.810±0.125	0.983±0.008
	Atten. U-Net [34]		0.863±0.052	0.713±0.145	5.861±2.335	0.853±0.087	0.836±0.119	0.987±0.007
	nnU-Net [35]		0.874±0.074	0.734±0.122	6.072±3.528	0.860±0.122	0.862±0.067	0.987±0.007
	UNETR [36]		0.874±0.090	0.710±0.149	6.394±4.770	0.857±0.129	0.870±0.072	0.988±0.006
	nnFormer [37]		0.879±0.100	0.744±0.127	6.430±6.231	0.858±0.136	0.861±0.082	0.988±0.005
	TransBTS [5]		0.881±0.102	0.743±0.114	4.922±3.756	0.858±0.138	0.873±0.052	0.988±0.006
	MultiCNN [38]		0.889±0.077	0.741±0.107	5.477±5.077	0.864±0.010	0.875±0.073	0.988±0.004
	MultiFormer [39]		0.871±0.092	0.716±0.123	6.650±5.693	0.868±0.130	0.871±0.080	0.987±0.023
	TuningUNet [40]		0.886±0.100	0.741±0.128	5.258±7.317	0.864±0.128	0.868±0.080	0.988±0.007
	Baseline-1		0.869±0.033	0.723±0.074	6.472±5.363	0.849±0.054	0.817±0.057	0.972±0.013
	Baseline-2		0.877±0.038	0.735±0.107	4.735±2.034	0.876±0.063	0.864±0.049	0.983±0.008
	Baseline-3		0.879±0.036	0.736±0.106	4.625±1.730	0.877±0.047	0.861±0.051	0.983±0.008
	Ours		0.897±0.037	0.768±0.076	4.287±1.814	0.909±0.055	0.879±0.050	0.989±0.006

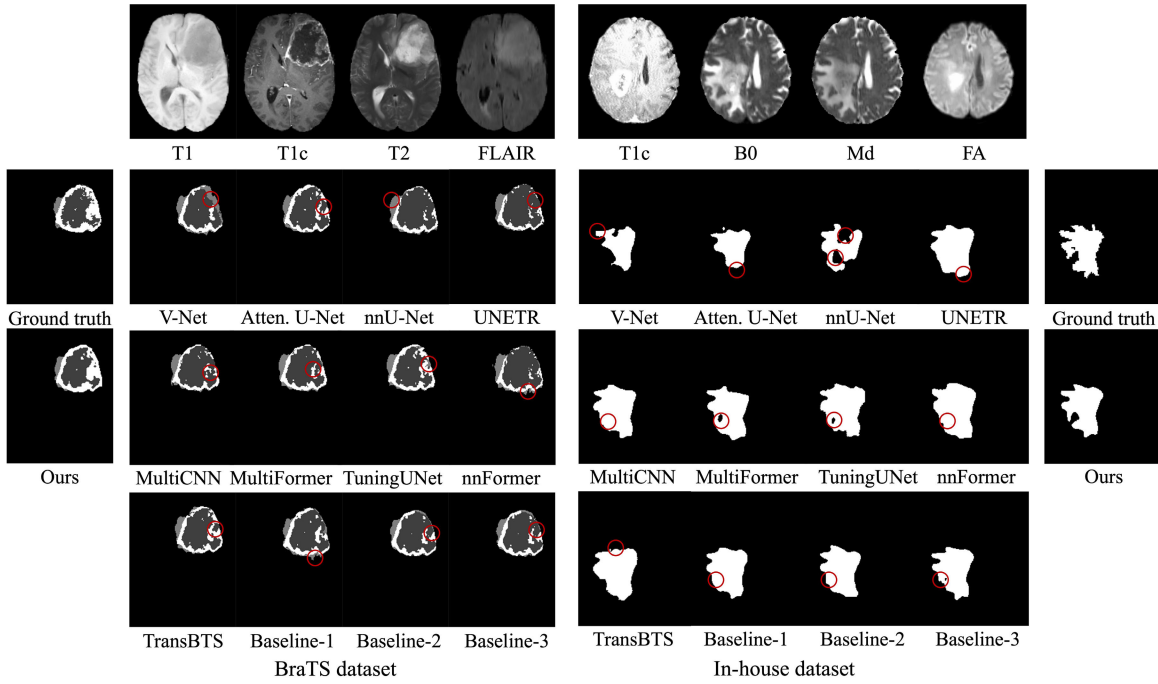


Fig. 5. Examples of segmentation results using all methods under evaluation in the BraTS2022 dataset (left) and in-house dataset (right). Note that, the BraTS2022 dataset contains three tumor ROIs, i.e., edema (gray), enhancing core (white), and necrosis (dark gray), while the in-house dataset has one whole tumor region. It is clear that the segmentation results using our framework are more accurate than the other methods under evaluation in both BraTS2022 and in-house datasets, especially at the regions marked in red circles.

the normal appearance network, $F_{T,l}$ learned from multimodal tumor brain images and $F_{R,l}$ learned from monomodal normal brain images suffer from the incomparable issue as no SimSiam network is applied before feature comparison. Therefore, the improvement using Baseline-3 is minor as compared with Baseline-2 which has no normal appearance network.

Fig. 5 shows some examples of the segmented brain tumors using each method under evaluation on both datasets. It is clear that for each tumor ROI in the BraTS2022 dataset and the whole tumor region in the in-house dataset, the segmentation results using our framework are more consistent with the ground truth than the other SOTA methods under evaluation, especially at the regions marked in red circles.

B. Evaluation of Global Correlation Block

Table II shows that the global correlation block (GCB) can do more effective fusion of features learned from multimodal images than the traditional cross modality convolution (CMC). As aforementioned (Section II-A), at each encoding level l of the segmentation backbone, GCB fuses features of different modalities channel-wisely according to the weighting vector $\omega_l^c \in \mathbb{R}^M$, $c = 1, \dots, C$ defined in (2). Each element in ω_l^c indicates the importance of the corresponding modality at the c -th channel. $C = 64, 128, 256$, and 512 at the encoding level $l = 1, 2, 3$, and 4 , respectively. Fig. 6 shows the average $\omega_l^c(m)$, i.e., average weight of the m -th modality, of all input images calculated at the testing stage. The vertical axis indicates the average weights of each modality, and the horizontal axis indicates the channel. Dotted lines indicate the quarter and three-quarter weights.

It is clear that for the BraTS2022 dataset, the weights of T2 and FLAIR are close to one at encoding level $l = 1$ and 2 ,

while the weights of T1 and T1c are close to zero. This is because T2 and FLAIR exhibit clear boundary and shape of whole brain tumor, which can be directly reflected by low-level features. As the network depth increasing (encoding level $l = 3$ and 4), details of tumor structure contained in T1 and T1c are learned as high-level features, thus, the weights of T1 and T1c are increased. For the in-house dataset, the trend of modality weights is similar to the BraTS2022 dataset. Specifically, B0, MD, and FA, which have high contrast tumor appearance, are assigned with large weights at low levels, while for T1c, which contains details of tumor structure, large weights are assigned at high levels.

C. Evaluation of Feature Alignment Module

To give an intuitive view of the effect of the feature alignment module (FAM) as well as the SimSiam network in it, distributions of features in $\zeta(F_{T,l})$ and $g(F_{R,l})$ (after feature alignment) are plotted in Fig. 7. Specifically, features, i.e., $\zeta(F_{T,l})(x)$ and $g(F_{R,l})(x)$, are divided into three classes: (1) tumor regions in $\zeta(F_{T,l})$; (2) normal regions in $\zeta(F_{T,l})$; and (3) normal regions in $g(F_{R,l})$. All features are projected onto a 2D plane using principal component analysis (PCA) [42].

Clearly, with the SimSiam network, features at normal regions in $\zeta(F_{T,l})$ and $g(F_{R,l})$ can be effectively aligned and are different from the features at tumor regions in $\zeta(F_{T,l})$. In contrast, good feature alignment cannot be achieved without the SimSiam network, causing the feature incomparable issue. Fig. 8 shows some examples of the attention maps A_l produced after the feature comparison at each decoding level ($l = 1, \dots, 4$) of our framework and Baseline-3. It is clear that, A_l of our framework has more concentrated tumor regions than

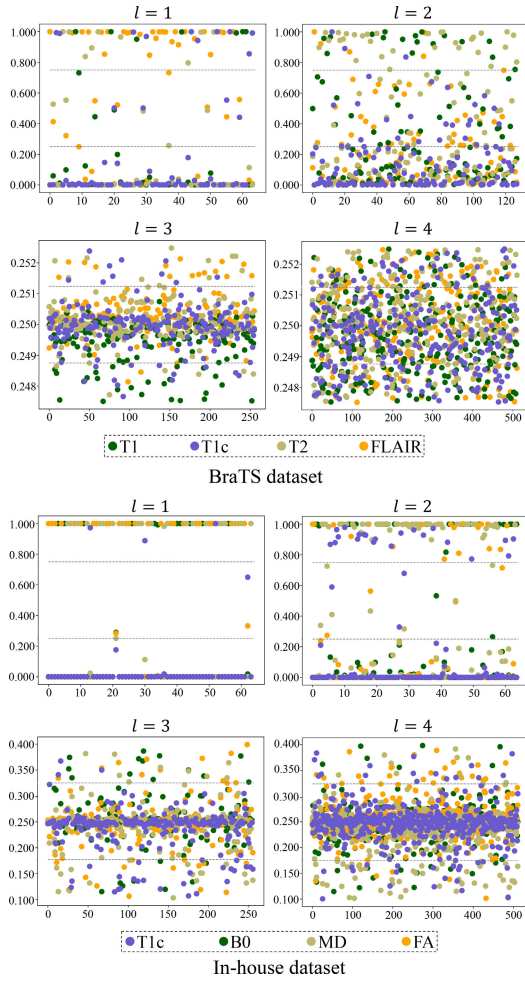


Fig. 6. Modality weights during the channel-wise feature fusion in the fusion module at each encoding level l of the segmentation backbone using the BraTS2022 and in-house datasets. The vertical axis represents weights, and the horizontal axis indicates channels.

Baseline-3. As a result, the final tumor segmentation results using our framework is more consistent with the ground truth.

D. Sensitivity to Normal Brain Images

Normal brain images, which are used as reference to high-light tumor-related features, play an important role in our framework. In this section, we evaluate the influence of normal brain images to our framework. Specifically, we conduct this experiment to answer the following questions: (1) How does image noise in the reconstructed normal brain images affect the segmentation? (2) What is the performance of our framework using other generative models instead of IntroVAE? (3) Can we use normal brain images from healthy subjects in our framework?

To the first question, we gradually add Gaussian noise to each normal brain image reconstructed by IntroVAE following a forward Markovian diffusion process [43] over T iterations:

$$q(I_R^T | I_R) = \prod_{t=1}^T q(I_R^t | I_R^{t-1}) \quad (8)$$

$$q(I_R^t | I_R^{t-1}) = \mathcal{N}(I_R^t; \sqrt{\alpha_t} I_R^{t-1}, (1 - \alpha_t) I), \quad (9)$$

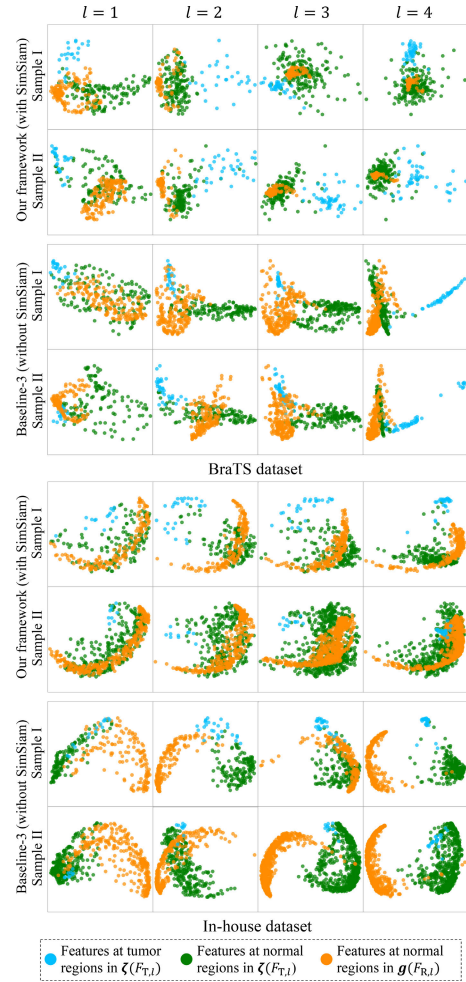


Fig. 7. Distribution of features in $\xi(F_{T,l})$ and $g(F_{R,l})$ at each decoding level l of our framework and Baseline-3 ($l = 1, \dots, 4$) using BraTS2022 and In-house datasets. Features are from three classes, i.e., tumor regions in $\xi(F_{T,l})$, normal regions in $\xi(F_{T,l})$ and normal regions in $g(F_{R,l})$, which are colored in blue, green and orange, respectively.

where I_R is the original normal brain image reconstructed by IntroVAE, q is data distribution, $\mathcal{N}$ represents Gaussian distribution, the scalar parameters $\alpha_{1:T}$ determine the variance of the noise added at each iteration, and I is the identity matrix. At each iteration $t = 1, \dots, T$, the distribution of image I_R^t is a combination of I_R^{t-1} and Gaussian noise, and α_t determines how much the information of I_R^{t-1} is preserved. Thus, the higher t , the lower quality of the resulting normal brain image I_R^t . In our experiment, $\alpha_{1:T}$ is set to linearly increased from $\alpha_1 = 0.001$ to $\alpha_T = 0.02$. We evaluate the segmentation results of our framework using $t = 200, 400, 700$, and 1000 , respectively. Details of the evaluation results are shown in Table III. It is clear that the performance of tumor region segmentation is degraded as the quality of normal brain images decreases. Specifically, for $t = 200$, the segmentation results is relatively stable, but as the t increases the performance is sharply decreased. Therefore, our framework is sensitive to the noises in normal brain images.

To the second question concerning generative models, our framework currently uses IntroVAE to reconstruct high quality normal brain images from corresponding tumor brain images as input of the normal appearance network. In this

TABLE III
SENSITIVITY OF OUR FRAMEWORK TO THE QUALITY OF NORMAL BRAIN IMAGES

Dataset	Method		Dice score	Jaccard index	HD	Sensitivity	Precision	Specificity
BraTS 2022	From IntroVAE with noise $t = 200$	ET	0.838±0.174	0.751±0.196	4.284±5.707	0.822±0.196	0.891±0.143	0.998±0.005
		TC	0.878±0.143	0.806±0.182	6.150±10.061	0.860±0.160	0.902±0.125	0.998±0.004
		WT	0.881±0.102	0.799±0.138	7.234±9.570	0.829±0.145	0.920±0.045	0.997±0.004
	From IntroVAE with noise $t = 400$	ET	0.803±0.182	0.701±0.201	4.378±5.396	0.757±0.206	0.700±0.142	0.995±0.004
		TC	0.845±0.154	0.756±0.190	7.418±9.500	0.799±0.179	0.730±0.121	0.988±0.015
		WT	0.815±0.130	0.705±0.162	8.553±8.242	0.724±0.170	0.768±0.140	0.932±0.013
	From IntroVAE with noise $t = 700$	ET	0.742±0.193	0.620±0.206	4.426 ±4.198	0.657±0.214	0.614±0.235	0.993±0.005
		TC	0.786±0.171	0.675±0.200	8.418±8.459	0.703±0.197	0.639±0.215	0.938±0.017
		WT	0.714±0.154	0.575±0.173	9.397±6.324	0.585±0.179	0.575±0.234	0.918±0.023
	From IntroVAE with noise $t = 1000$	ET	0.662±0.201	0.525±0.202	4.489±3.504	0.546±0.210	0.527±0.229	0.992±0.013
		TC	0.710±0.188	0.579±0.204	8.718±7.716	0.595±0.204	0.549±0.210	0.926±0.025
		WT	0.601±0.164	0.448±0.163	9.537±2.879	0.453±0.167	0.581±0.228	0.887±0.014
	From VAE [13]	ET	0.852±0.163	0.768±0.190	4.007±7.972	0.858±0.184	0.878±0.138	0.998±0.004
		TC	0.884±0.151	0.818±0.189	5.967±10.501	0.884±0.167	0.912±0.124	0.998±0.004
		WT	0.912±0.087	0.847±0.116	6.931±11.345	0.897±0.118	0.937±0.067	0.996±0.004
	From f-AnoGAN [44]	ET	0.790±0.180	0.683±0.201	4.501±4.701	0.743±0.206	0.889±0.137	0.997±0.004
		TC	0.827±0.151	0.729±0.188	8.047±8.702	0.773±0.179	0.912±0.120	0.996±0.004
		WT	0.829±0.113	0.723±0.147	9.117±7.893	0.753±0.154	0.939±0.052	0.997±0.004
	From healthy subjects	ET	0.856±0.168	0.775±0.192	3.550±6.308	0.862±0.188	0.879±0.146	0.998±0.004
		TC	0.894±0.137	0.829±0.177	5.543±10.535	0.896±0.148	0.912±0.128	0.998±0.004
		WT	0.918±0.072	0.855±0.106	6.248±10.924	0.905±0.106	0.941±0.054	0.996±0.005
	Ours	ET	0.869±0.166	0.780±0.191	3.323±5.785	0.865±0.186	0.893±0.139	0.998±0.004
		TC	0.897±0.134	0.834±0.175	5.118±10.104	0.898±0.147	0.921±0.139	0.998±0.004
		WT	0.929±0.066	0.868±0.099	5.844±10.469	0.914±0.100	0.945±0.052	0.997±0.003
In-house	From IntroVAE with noise $t = 200$		0.877±0.075	0.743±0.101	10.037±10.047	0.894±0.078	0.830±0.071	0.983±0.008
	From IntroVAE with noise $t = 400$		0.819±0.128	0.668±0.141	52.399±20.078	0.889±0.110	0.728±0.145	0.970±0.009
	From IntroVAE with noise $t = 700$		0.705±0.149	0.532±0.143	65.064±12.070	0.873±0.083	0.566±0.149	0.937±0.012
	From IntroVAE with noise $t = 1000$		0.693±0.148	0.525±0.144	65.424±11.206	0.861±0.081	0.564±0.153	0.937±0.014
	From VAE [13]		0.869±0.093	0.735±0.121	21.290±21.559	0.909±0.084	0.827±0.119	0.985±0.008
	From f-AnoGAN [44]		0.777±0.138	0.720±0.146	24.754±13.762	0.890±0.078	0.764±0.156	0.962±0.011
	From healthy subjects		0.889±0.033	0.763±0.085	4.259±1.839	0.890±0.045	0.864±0.046	0.980±0.006
	Ours		0.897±0.037	0.768±0.076	4.287±1.814	0.909±0.055	0.879±0.050	0.989±0.006

experiment, we replace IntroVAE with other generative models commonly adopted in anomaly detection methods, such as VAE [13] and f-AnoGAN [44]. The training dataset for each generative model is the same as that used in IntroVAE, i.e., the IXI dataset. Some examples of the normal brain images reconstructed by IntroVAE, VAE, and f-AnoGAN are shown in Fig. 9. It is clear that the reconstructed normal brain images using VAE are highly blurry, and important anatomical structure, such as the brain sulcus and gyrus, is missing. f-AnoGAN can reconstruct relatively clear brain anatomical structure, but it cannot preserve anatomical consistency well. Since IntroVAE combines the merits of both VAE and GAN, the reconstruction quality is better than VAE and f-AnoGAN. In our experiment, VAE and f-AnoGAN instead of IntroVAE are respectively used in our framework to produce the normal brain images for the normal appearance network. Details of the evaluation results are shown in Table III. Clearly, the final segmentation results using VAE and f-AnoGAN are inferior to IntroVAE. Interestingly, the evaluation results show that our framework with VAE is better than f-AnoGAN, the main reason is that although the reconstructed normal brain images using f-AnoGAN is clear, the anatomical structure is usually

unreal and disordered. In contrast, the anatomical structure using VAE is consistent with human brain in general.

To the third question about using normal brain images from healthy subjects instead of reconstruction. For each input multimodal tumor brain images of our framework, T1 MR normal brain images of healthy subjects, which have the minimal L2 distance to the T1 modality of the tumor brain images, are selected from the IXI normal brain dataset as the input of the normal appearance network. Clearly, because of inter-subject variation, normal brain images of healthy subjects usually have lower anatomical consistency with tumor brain images than those reconstructed by IntroVAE. To quantify such consistency, average image intensity error ratio (ER) between each tumor brain image and its corresponding normal brain image is calculated, which is defined as

$$ER(I_T, I_R) = \frac{\sum_{x \in \Omega_{NR}} |I_T(x) - I_R(x)|}{\sum_{x \in \Omega_{NR}} I_T(x)}, \quad (10)$$

where I_T and I_R indicate tumor brain image (T1 modality) and its corresponding T1 MR normal brain image (reconstructed by IntroVAE or from healthy subjects), respectively, and Ω_{NR} stands for normal regions in I_T . Low ER indicates high

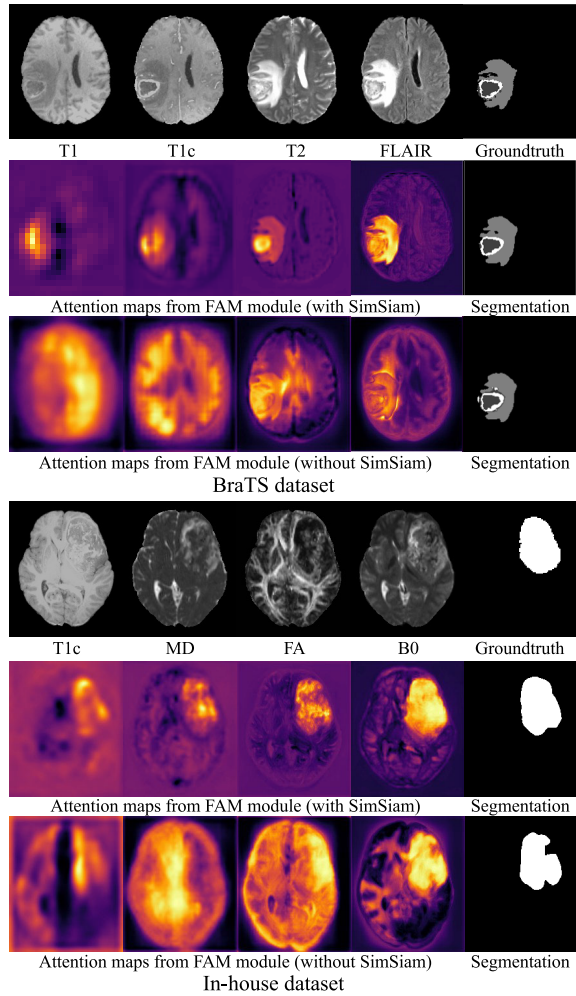


Fig. 8. Examples of the attention maps generated after the feature comparison at each decoding level. Clearly, the quality of the attention maps and the final segmentation results produced by our framework (with SimSiam) are better than Baseline-3 (without SimSiam).

consistency between I_T and I_R . For the BraTS2022 dataset, the averages and standard deviations of the error ratio are 0.274 ± 0.043 (IntroVAE) vs. 0.495 ± 0.168 (healthy subjects), and for the in-house dataset, the results are 0.378 ± 0.010 (IntroVAE) vs. 0.573 ± 0.018 (healthy subjects). The corresponding p values of Wilcoxon signed rank test are 1.120×10^{-12} (BraTS2022) and 1.149×10^{-6} (in-house dataset), respectively. As aforementioned, for each input tumor brain images, the most similar T1 MR normal brain image from healthy subject is selected from the IXI dataset and sent to the normal appearance network. The evaluation results of both BraTS2022 and in-house datasets are shown in Table III. It can be seen that the performance of our framework is little decreased but still comparable to that using IntroVAE.

E. Evaluation of Different Segmentation Backbone

The segmentation backbone of our framework is of a simple U-net like encoder-decoder structure. Actually, the segmentation backbone can be replaced by other segmentation networks, such as TransBTS [5]. In this section, we evaluate the generalization ability of our framework and show the advantage of integrating normal appearance network.

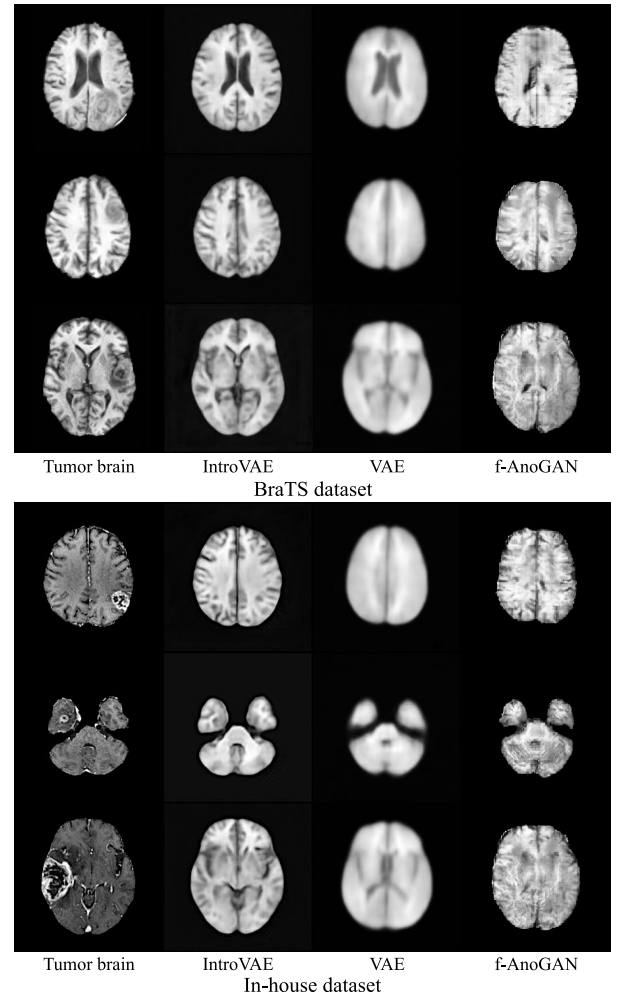


Fig. 9. Examples of reconstructed normal brain images of BraTS2022 and in-house datasets using IntroVAE, VAE, and f-AnoGAN. Specifically, VAE has blurred reconstruction structure, f-AnoGAN has clear but disordered structure, IntroVAE has clear and reasonable structure.

Specifically, we adopt TransBTS as the segmentation backbone of our framework and compare with pure TransBTS on the BraTS2022 and in-house datasets. The evaluation results are shown in Table IV. It is clear that TransBTS in our framework, which is denoted as Ours (TransBTS), achieves better performance than pure TransBTS in most metrics except the Specificity on the in-house dataset. Therefore, our framework, which uses normal brain images as reference to conduct feature-level comparison with tumor brain images, is broadly applicable and can boost the segmentation performance of existing methods.

F. Model Complexity

To make a comprehensive comparison of all methods under evaluation, besides the segmentation accuracy reported in the previous sections, network parameters, number of floating point operations (FLOPs), and average inference time are investigated. The evaluation results are shown in Table V. Specifically, except V-Net, and baseline-1 and -2 from the ablation study, our framework has the smaller number of parameters than the rest methods. The FLOPs of our framework is relatively large, and the main reason is the global

TABLE IV
EVALUATION RESULTS OF OUR FRAMEWORK USING TRANSBTS AS SEGMENTATION BACKBONE

Dataset	Method		Dice score	Jaccard index	HD	Sensitivity	Precision	Specificity
BraTS 2022	TransBTS [5]	ET	0.850±0.177	0.770±0.200	4.330±9.708	0.844±0.201	0.892±0.139	0.994±0.003
		TC	0.873±0.189	0.810±0.214	6.043±11.971	0.859±0.205	0.921±0.146	0.996±0.005
		WT	0.913±0.081	0.849±0.115	9.096±16.448	0.913±0.104	0.921±0.079	0.996±0.004
	Ours (TransBTS)	ET	0.859±0.157	0.779±0.187	3.427±6.099	0.858±0.181	0.898±0.138	0.998±0.004
		TC	0.889±0.157	0.827±0.192	4.883±9.100	0.882±0.175	0.927±0.117	0.998±0.005
		WT	0.926±0.087	0.864±0.119	5.456±9.254	0.916±0.102	0.948±0.053	0.996±0.005
In-house	TransBTS [5]		0.881±0.102	0.743±0.114	4.922±3.756	0.858±0.138	0.873±0.052	0.988±0.006
	Ours (TransBTS)		0.896±0.050	0.767±0.073	4.916±1.422	0.890±0.071	0.887±0.045	0.984±0.009

TABLE V
EVALUATION RESULTS OF MODEL COMPLEXITY

Model	Params (M)	FLOPs (G)	Inference time (s)
V-Net [33]	9.36	6.87	0.089
Atten. U-Net [34]	34.88	31.26	0.135
nnU-Net [35]	64.14	59.53	0.159
UNETR [36]	90.86	12.86	0.208
nnFormer [37]	89.29	12.41	0.198
TransBTS [5]	24.23	12.84	0.117
MultiCNN [38]	177.31	91.01	0.239
MultiFormer [39]	89.87	12.72	0.122
TuningUNet [40]	59.12	27.65	0.188
Baseline-1	14.50	29.77	0.181
Baseline-2	17.62	31.06	0.166
Baseline-3	19.92	35.22	0.179
Ours	19.92	35.22	0.179
Ours (TransBTS)	26.16	15.14	0.159

correlation block (GCB), which is used for multimodal fusion and requires large FLOPs. In a similar way, Baseline-1, which adopt cross modality convolution (CMC) for multimodal fusion, require large FLOPs too. It is worth noting that all methods in our experiment take 2.5D slices as input, therefore, the network parameter, FLOPs, and average inference time are smaller than those working on 3D.

IV. CONCLUSION

In this paper, we proposed a novel deep learning framework for multimodal brain tumor segmentation, where monomodal normal brain images were used as reference to compare with multimodal tumor brain images in a learned feature space. In this way, tumor-related features can be highlighted and enhanced for good segmentation. In our framework, a new self-attention based global correlation block (GCB) was presented to capture global correlations among features learned from different modalities of tumor brain images for effective feature fusion. Moreover, a new feature alignment module (FAM) was proposed to tackle the incomparable issue between features learned from multimodal tumor brain images and monomodal normal brain images. Based on FAM, effective feature comparison can be achieved, and high quality attention

maps A_l can be generated, finally boosting the performance of brain tumor segmentation.

Both public BraTS2022 and in-house datasets were used to evaluate our framework. The experimental results showed that our framework achieved the best performance of all methods under evaluation in terms of Dice score, Sensitivity, Hausdorff distance, Precision, Specificity, and Jaccard index. Besides the comparison with the state-of-the-art segmentation methods, ablation experiments were also conducted and demonstrated that all proposed components, including IntroVAE, GCB, FAM, and the normal appearance network, have positive impacts on the segmentation results. Moreover, we investigated the performance of our framework using normal brain images with different qualities and sources. The experimental results showed that our framework can be influenced by the noise in normal brain images and the performance is relatively stable using normal brain images from healthy subjects. In addition, the evaluation results using VAE and f-AnoGAN showed that reconstructed normal brain images with blurred but consistent anatomical structure in general can provide more effective information to enhance the segmentation than clear but disordered anatomical structure. At last, we evaluated our framework using different segmentation backbone, and the results showed that our framework has high generalization ability and can improve the performance of existing segmentation networks.

For the future study, although IntroVAE and FAM with SimSiam are used in our framework, we noted that the domain gap between the monomodal normal brain images and the multimodal tumor brain images is still non-negligible. Thus, we will explore possible ways of getting multimodal normal brain images. Moreover, our framework has relatively large computational workload, and we will explore more lightweight techniques for effective fusion of cross-modality.

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